

Newton's Second Law

Newton's Second Law

- A body **accelerates** when acted upon by a **net external force**.
- The acceleration is proportional to the net force and is in the direction which the net force acts.

Newton's Second Law

- $\Sigma F = ma$

- where ΣF is the net force measured in Newtons (N)
- m is mass (kg)
- a is acceleration (m/s^2)

Working 2nd Law Problems

1. Draw a force or free body diagram.
2. Set up 2nd Law equations in each dimension.
 $\Sigma F_x = ma_x$ and/or $\Sigma F_y = ma_y$
3. Identify numerical data.
x-problem and/or y-problem
4. Solve the equations.
5. Substitute numbers into equations.
“plug-n-chug”

Sample Problem

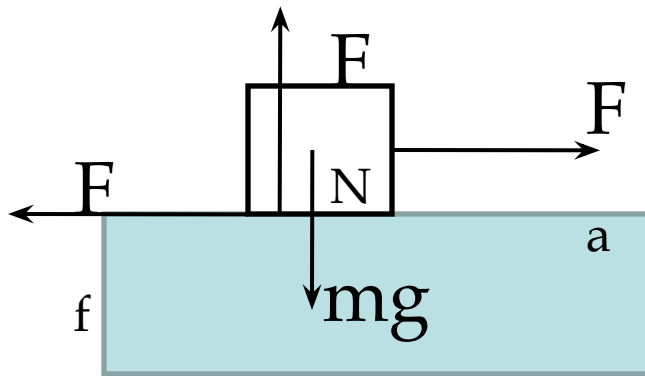
In a grocery store, you push a 14.5-kg cart with a force of 12.0 N. If the cart starts at rest, how far does it move in 3.00 seconds?

Sample problem

Suppose a crane accelerates a 1500 kg crate upward at 1.2 m/s^2 . What is the tension in the cable?

N.S.L

A 10-kg box is being pulled across the table to the right by a rope with an applied force of 50N. Calculate the acceleration of the box if a 12 N frictional force acts upon it.



In which direction,
is this object
accelerating?

The X direction!

So N.S.L. is worked
out using the forces
in the “x” direction
only

Newton's Laws in 2D

Newton's 2nd Law in 2-D

- The situation is more complicated when forces act in more than one dimension.
- You must still identify all forces and draw your force diagram.
- You then resolve your problem into an x-problem and a y-problem (remember projectile motion????).

Sample problem

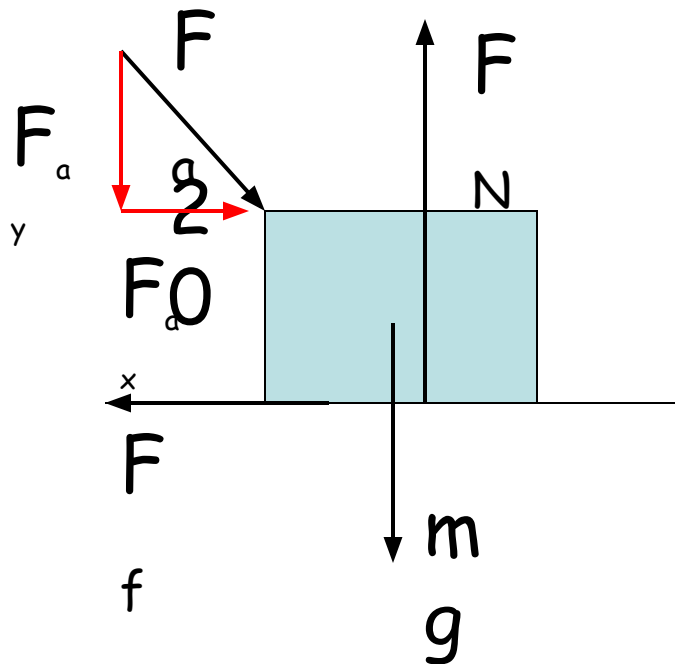
Larry pushes a 200 kg block on a frictionless floor at a 45° angle below the horizontal with a force of 150 N while Moe pulls the same block horizontally with a force of 120 N.

- a) Draw a free body diagram.
- b) What is the acceleration of the block?
- c) What is the normal force exerted on the block?

Example

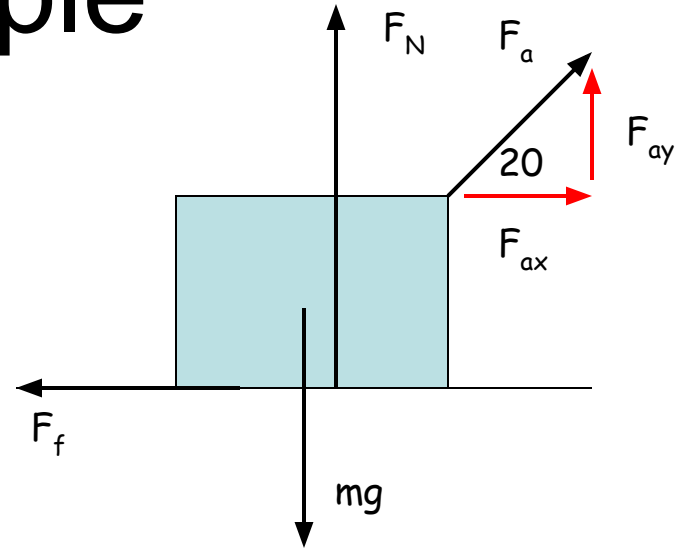
A 1500 N crate is being pushed across a level floor at a constant speed by a force F of 600 N at an angle of 20° below the horizontal as shown in the figure.

a) What is the coefficient of kinetic friction between the crate and the floor?



Example

A 600 N force is pulling a 1500 N block at an angle of 20° above the horizontal as shown in the figure, what will be the acceleration of the crate. The coefficient of friction is the 0.3.



Apparent weight

- If an object subject to gravity is not in free fall, then there must be a **reaction force** to act in opposition to gravity.
- We sometimes refer to this reaction force as **apparent weight**.

Elevator rides

- When you are in an elevator, **your actual weight (mg) never changes.**
- You feel lighter or heavier during the ride because your **apparent weight** increases when you are accelerating up, decreases when you are accelerating down, and is equal to your weight when you are not accelerating at all.

Going Up?

$$v = 0$$
$$a = 0$$

$$v \geq 0$$
$$a > 0$$

$$v > 0$$
$$a = 0$$

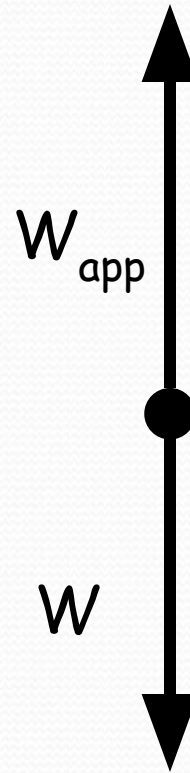
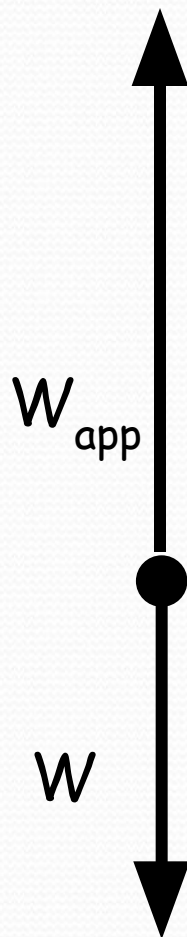
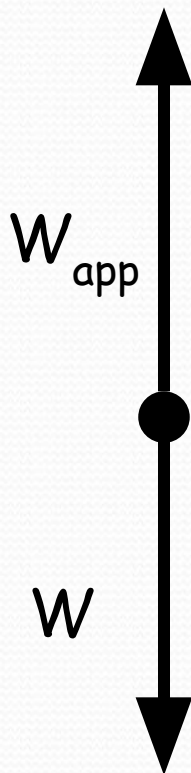
$$v \geq 0$$
$$a < 0$$

Heavy feeling

Normal feeling

Normal feeling

Light feeling



Ground floor

Just starting up

Between floors

Arriving at top floor

Going

Down?

$$v = 0$$

$$a = 0$$

$$v \leq 0$$

$$a < 0$$

$$v < 0$$

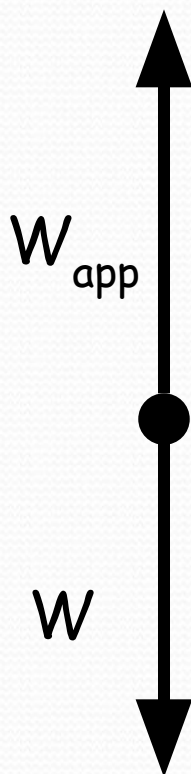
$$a = 0$$

$$v \leq 0$$

$$a > 0$$

Heavy feeling

Normal feeling



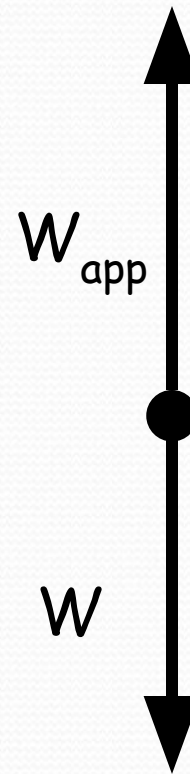
Top
floor

Light feeling

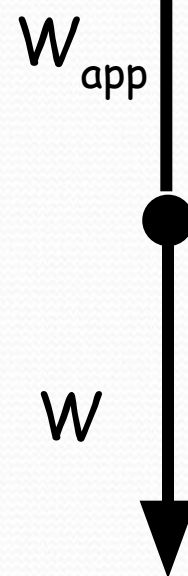


Beginning
descent

Normal feeling



Between
floors



Arriving at
Ground floor



Sample Problem

An 85-kg person is standing on a bathroom scale in an elevator. What is the person's apparent weight

- a) when the elevator accelerates upward at 2.0 m/s^2 ?
- b) when the elevator is moving at constant velocity between floors?
- c) when the elevator begins to slow at the top floor at 2.0 m/s^2 ?

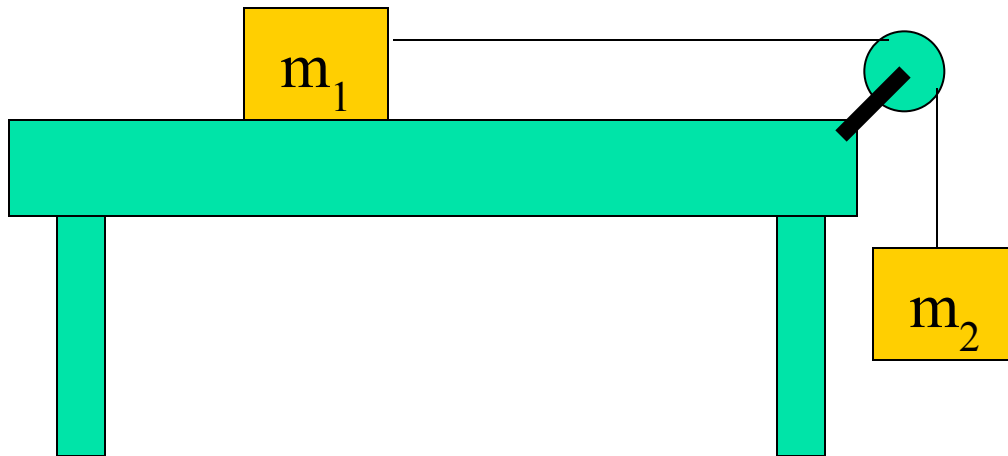
Sample Problem

How long will it take a 1.0 kg block initially at rest to slide down a frictionless 20.0 m long ramp that is at a 15° angle with the horizontal?

Sample problem

Mass 1 (10 kg) rests on a table connected by a string to Mass 2 (5 kg). If $\mu_s = 0.30$ and $\mu_k = 0.20$, what is

- (a) the acceleration of each block?
- (b) the tension in the connecting string?



Sample problem - solution (a)

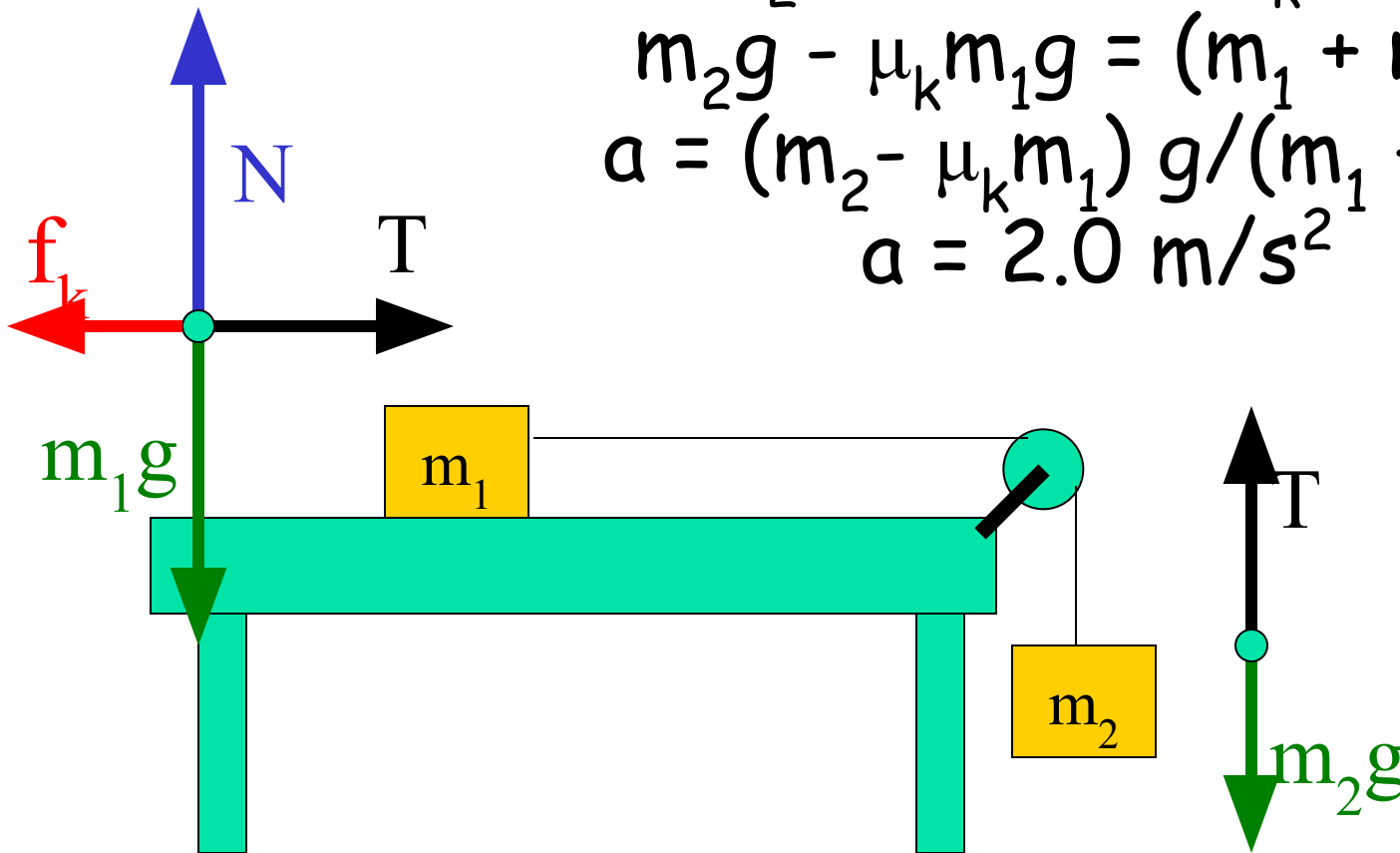
$$\Sigma F = ma$$

$$m_2g - T + T - f_k = ma$$

$$m_2g - \mu_k m_1g = (m_1 + m_2)a$$

$$a = (m_2 - \mu_k m_1)g / (m_1 + m_2)$$

$$a = 2.0 \text{ m/s}^2$$



Using that the acceleration is 2.0 m/s^2 from part a)

Sample problem - solution (b)

Using block 2

$$\Sigma F = ma$$

$$m_2 g - T = m_2 a$$

$$T = m_2 (g - a)$$

$$T = 40 \text{ N}$$

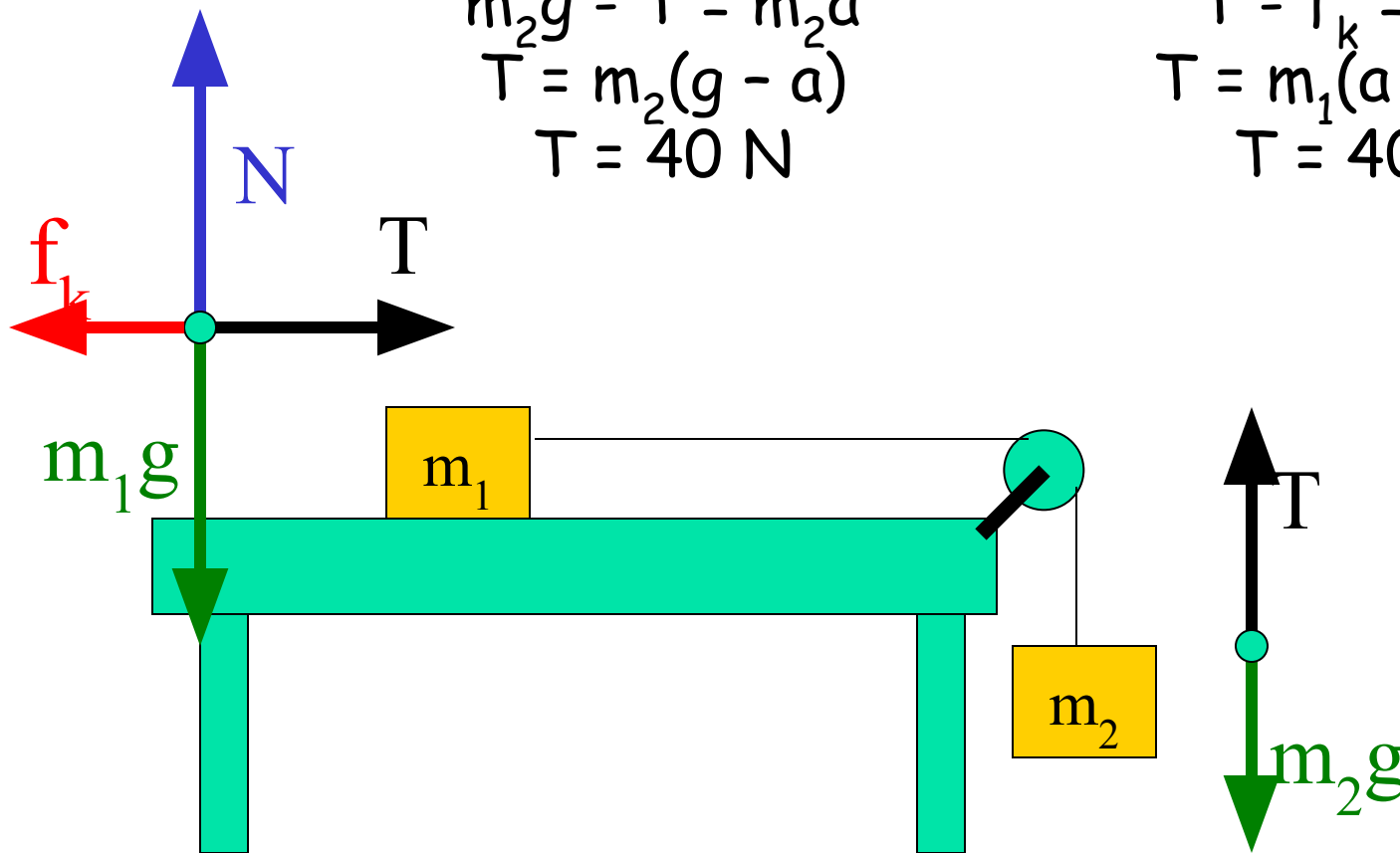
Using block 1

$$\Sigma F = ma$$

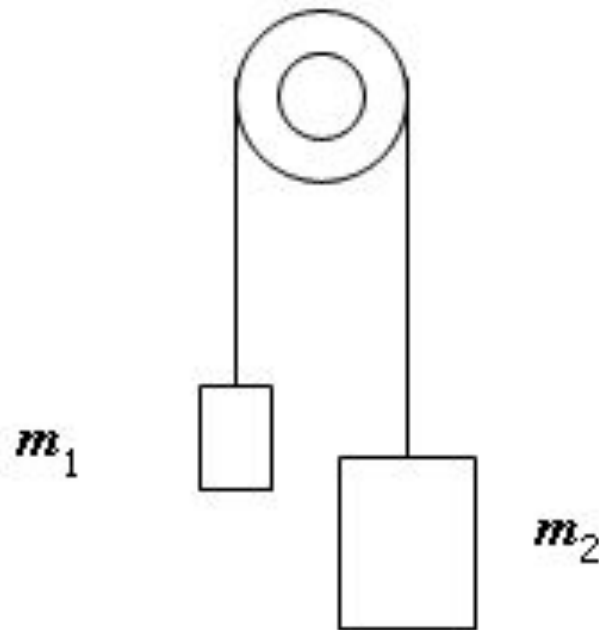
$$T - f_k = m_1 a$$

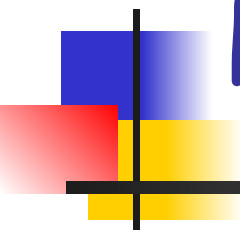
$$T = m_1 (a + \mu_k g)$$

$$T = 40 \text{ N}$$



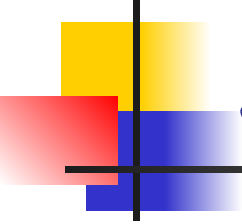
An Atwood's machine has $m_1 = 1\text{kg}$, $m_2 = 2\text{kg}$, hung from an ideal pulley. What is the acceleration of the masses? Calculate the tension in the string attached to each mass.





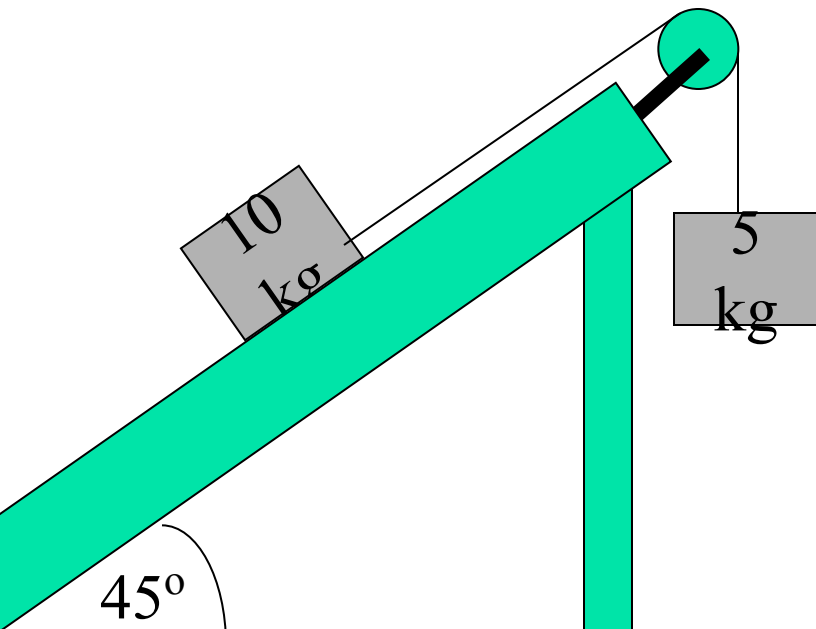
Pulleys and Ramps - together

Ramps and Pulleys - together!



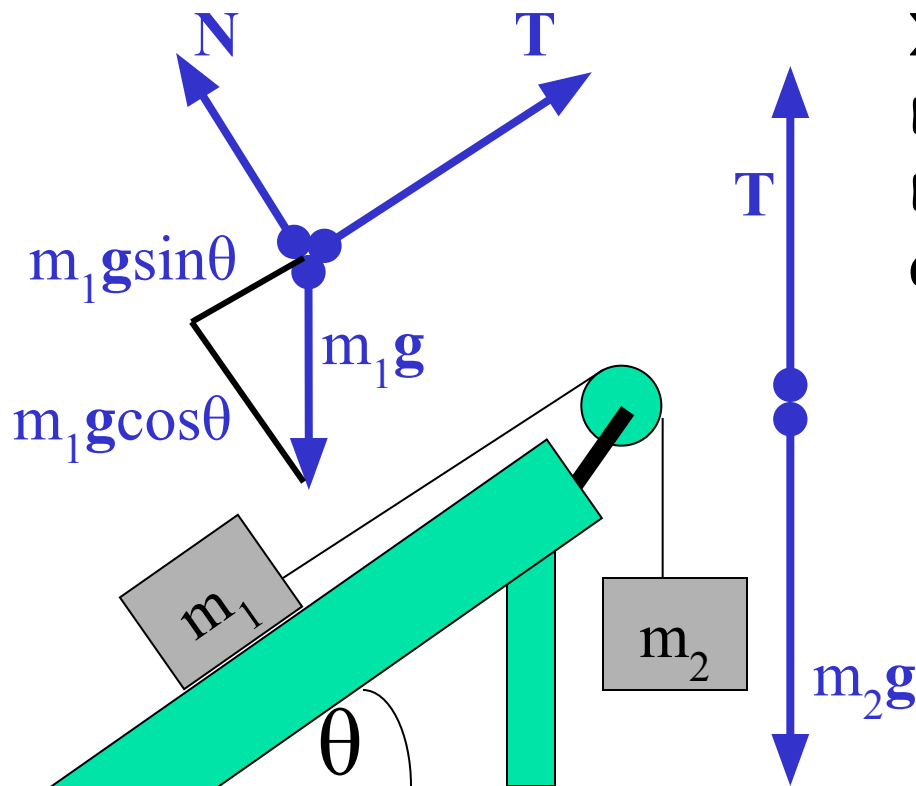
Sample problem

- Two blocks are connected by a string as shown in the figure. What is the acceleration, assuming there is no friction?



Magic pulleys on a ramp

- It's a little more complicated when a magic pulley is installed on a ramp.



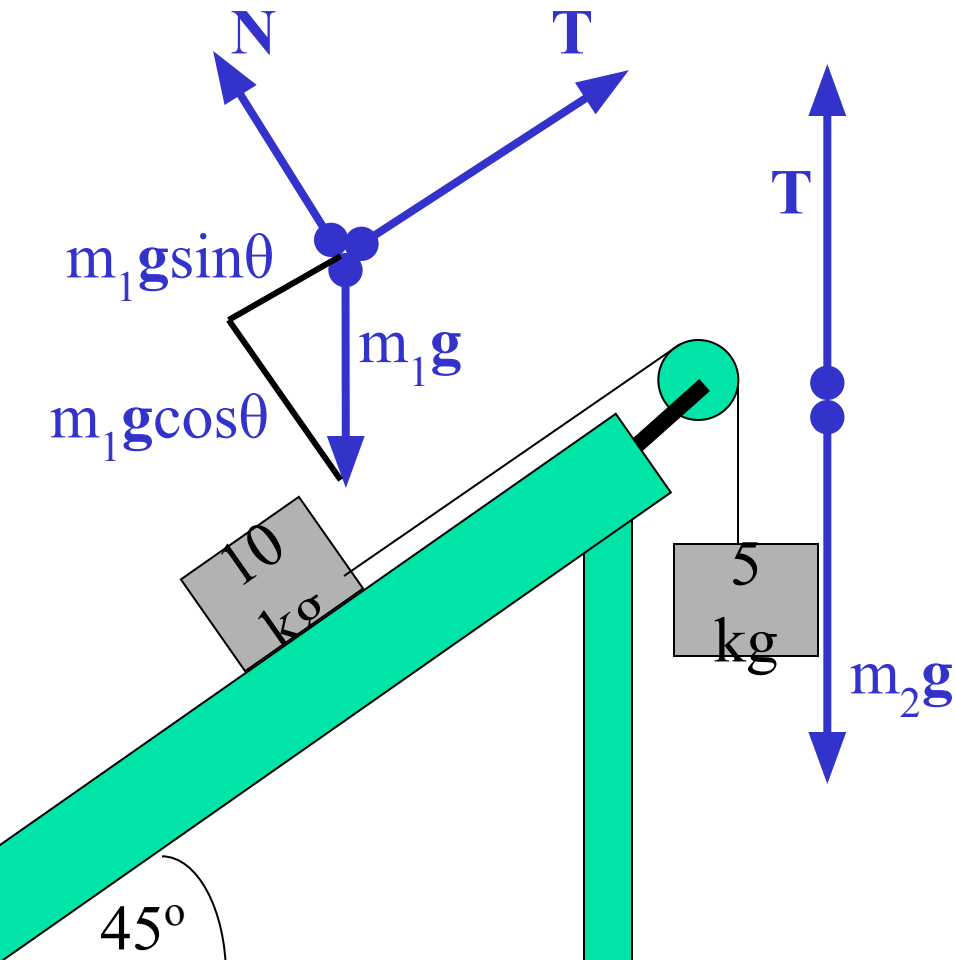
$$\Sigma \mathbf{F} = m\mathbf{a}$$

$$m_2 g - T + T - m_1 g \sin \theta = (m_1 + m_2) a$$

$$m_2 g - m_1 g \sin \theta = (m_1 + m_2) a$$

$$a = (m_2 - m_1 \sin \theta) g / (m_1 + m_2)$$

Sample problem - solution



$$\Sigma \mathbf{F} = m\mathbf{a}$$

$$m_2 g - T + T - m_1 g \sin \theta = (m_1 + m_2) a$$

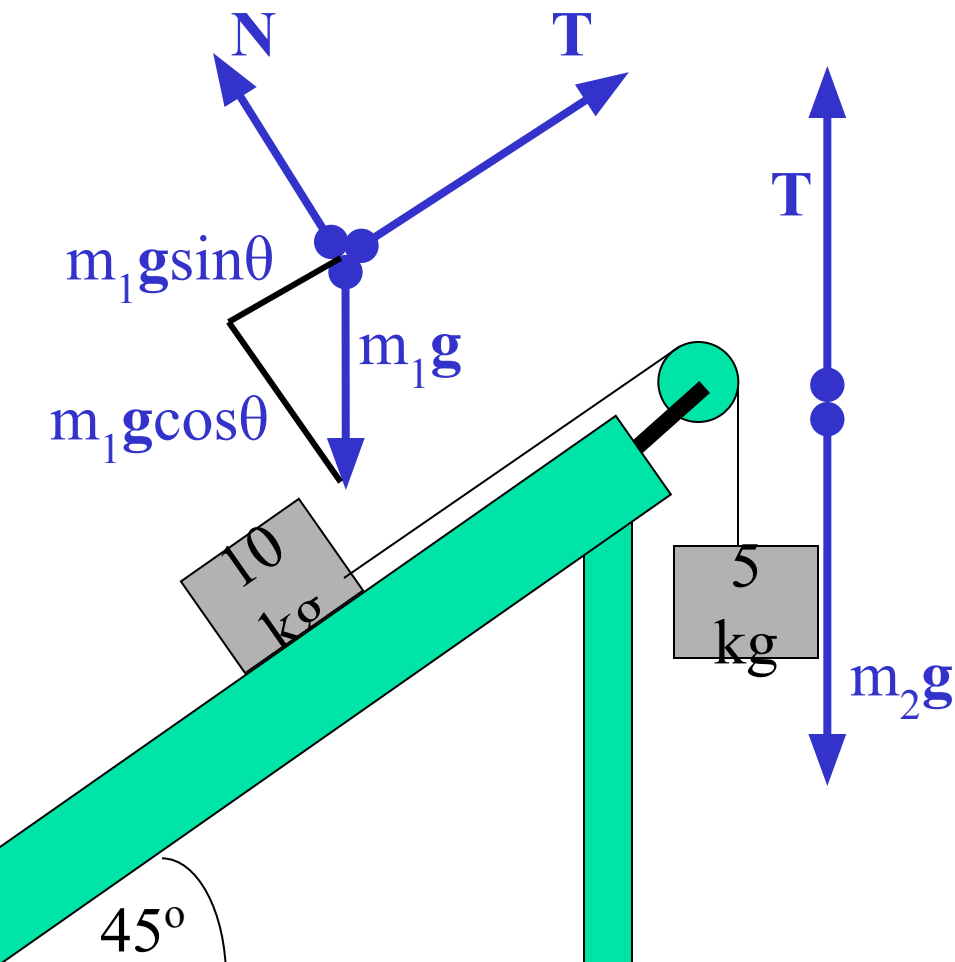
$$m_2 g - m_1 g \sin \theta = (m_1 + m_2) a$$

$$a = (m_2 - m_1 \sin \theta) g / (m_1 + m_2)$$

$$a = [(5 - 10 \sin 45^\circ)(9.8)] / 15$$

$$a = -1.35 \text{ m/s}^2$$

Sample problem - solution



$$\Sigma \mathbf{F} = m\mathbf{a}$$

$$m_2 g - T + T - m_1 g \sin \theta = (m_1 + m_2) a$$

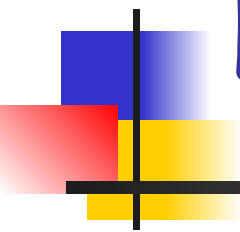
$$m_2 g - m_1 g \sin \theta = (m_1 + m_2) a$$

$$a = (m_2 - m_1 \sin \theta) g / (m_1 + m_2)$$

$$a = [(5 - 10 \sin 45^\circ)(9.8)] / 15$$

$$a = -1.35 \text{ m/s}^2$$

How would this change if there is friction on the ramp?



Newton's Laws

Applications

Problem

A 10-kg wooden box rests on a wooden ramp. The coefficient of static friction is 0.50, and the coefficient of kinetic friction is 0.30. What is the friction force between the box and ramp if

-) the ramp is at a 25° angle?
-) the ramp is at a 45° angle?
-) what is the acceleration of the box when the ramp is at 45° ?